Proceedings of the Seventh International Conference on Communication and Electronics Systems (ICCES 2022)
IEEE Xplore Part Number: CFP22AWO-ART; ISBN: 978-1-6654-9634-6

Analysis and Prognosis of Covid-19 using Machine Learning Algorithms and Visualization using Tableau

Aravind Swaminathan
Department of Electronics and Communication Engineering, Chennai Institute of Technology
Chennai, India
aravindsswaami@gmail.com

Surendran R
Department of Computer Science and Engineering
Saveetha School of Engineering
Saveetha Institute of Medical and Technical Sciences,
Chennai, India
surendran.phd.it@gmail.com

Joel Benjamin J
Department of Electronics and Communication Engineering, Chennai Institute of Technology
Chennai, India
joelbenjamin1990@gmail.com

Swaminathan Thiyagarajan
Digital Transformation Consultant, Tata Consultancy Services,
Chennai, India
swaami1969@gmail.com

2022 7th International Conference on Communication and Electronics Systems (ICCES) | 978-1-6654-9634-6/22/$31.00 ©2022 IEEE | DOI: 10.1109/ICCES54183.2022.9835941

Abstract*- This paper uses prognosticative machine learning models that predict corona positives and deaths as a result of the crisis, and the recovery rate from the pandemic. This method aids in diagnosing the contours of an individual's presumption in data transmission based on medical knowledge and calculates the unfolding virus's socio-economic impact. It examines the Covid-19's spread technique with the help of machine learning models. It also identifies the approaching prophecy and recessive presumption of the crisis at the same time, and as a result, this applicable analysis aids similar countries in making decisions. This paper also considers the global prevalence of the plague. Within the first phase of the irruption, eight supervised classification epidemiologic models are used to estimate the day-to-day and monomer incidents of coronavirus throughout the world, as well as the vital replica variety, growth rate, and increasing time. Calculations are also made for the more intricate efficacious replica variety, which reveals that since the predominant cases are confirmed to the specific countries, the severity has decreased. Machine learning models' prognosticative capabilities are found to provide an additional satisfactory match, and simple estimates of daily incidents around the world.

Key Words: Data Analysis, Data Cleaning, Machine Learning, Prediction, Classification Techniques, Supervised Learning.

I. INTRODUCTION

The coronavirus, also renowned as the Covid crisis, is a worldwide epidemic caused by a chronic respiratory infection called Covid-2, also known as the SARS virus. The original virus emerged in the Chinese city of Wuhan in December 2019 and attempts to contain it failed, allowing it to spread across the world. This flu epidemic, which began on March 13, 2022, claimed the lives of 456 million people and resulted in 6.04 million premature deaths, making it one of the deadliest in history.

Symptoms of the coronavirus can range from mild to fatal. The majority of people, on the other hand, suffer from a combination of chills, hacking, and exhaustion. The prevalence of the symptoms is more common in the elderly and those who have certain core illnesses. Inhaling air contaminated with pathogen-carrying drips and highly pathogenic particles is especially dangerous not only when persons are close to each other, but also when they're breathed in over longer distances. Contaminated liquids may also enter the iris, mouth, or nasal passages. Moreover, in rare instances, people can be infected through dirty surfaces. Covid patients are normally infectious for ten days and will transmit the disease irrespective of showing any symptoms. The changes have resulted in a plethora of strains (variations) with different degrees of infectiousness and pathogenicity. Coronavirus immunoglobulin's have been prolonged and significantly increased in several countries since December 2020. Social isolation, expanding ventilation, and alienating individuals who have been exposed are examples of additional recommended preventive measures. Monoclonal antibodies, novel antiviral agents, and side effect strategic planning are among the medications available.

Organizational interventions include restrictions on travel, curfews, professional constraints and terminations, work fulfilment risks, isolations, challenging structures, and following links of affected individuals. The pandemic wreaked havoc on society and the economy, with the worst global effects. Cache network disruptions and irrational acquiring were to be blamed for the overall primary commodities shortages, including food poverty. There was a noticeable drop in contamination as a result of the worldwide economic lockdowns. Educational institutions and public spaces were partial or completely closed in enough areas, and multiple events were cancelled or delayed.

Political pressures grew as duplicity flowed through electronic platforms and broad transmissions. The pandemic exacerbated racial and geographic segregation, as well as a lack of alignment between broad sense well-being goals, individual privileges, and well-being values. These social clubs are active on a global, provincial, and national level to ensure that those who have been impacted have a voice and were included in the response. Confirming that global proposals and connections are tested and adjusted to contexts will help governments gain sufficient control over pandemic outbreaks.

This paper proposes an integrated framework to deal with health data that community health practitioners can use to manage epidemics and develop health reactions and guidelines. This research focuses on the protagonists in the advancement of risk evidence broadcasting and investigating the behaviors that provide information intensification within a week of hearing the news.

II.EXISTING SYSTEM

A one-centre retrospective analysis of patient records from Wuhan's Jinyintan Hospital was performed. Though this survey cannot directly address coronavirus prognosis, it does provide a better understanding of the scientific implications. In China, radiographic shifts in Computed Tomography images of patients who had died from the coronavirus were discovered. Deep learning methods were used to drag Covid's explicit characteristics through scan images as a substitute analytical procedure. They gathered scan images of patients with positive corona as well as those with pneumonia. The outcomes of their work provided solid foundations for using intelligence in real-world coronavirus prediction. In contrast to this proposed work, which uses medical character traits and laboratory results to predict outcomes, the previous study used computed tomography images. Then, clinical centres in China defined the etiologic, systematic, testing facility, radiological, and therapy data. The data were studied and found to track infections.

To conduct this research, the following steps are considered:

- Keyword Categorization: The following keywords have been identified: Supervised Classification algorithms, Covid-19, Classification, Prediction, and Projection.
- Creating the Quest Lines: In order to develop the hunt string, prior keywords are assigned to the above-identified keywords.
- Finding the Literature: A series of searches are conducted on various digital database media, including “Google Scholars, IEEE, and Science Direct.”
- Following the evaluation criteria for selecting: The insertion and elimination criteria are applied to the gathered publications, such as tutelages and research papers, to limit the research.

Using in-built smart sensors, Halgurd S. Maghdid proposed a new technique for monitoring coronavirus disease. The AI framework gathered data from a variety of sensors to forecast the expenses of pneumonia, as well as the disease's contamination [1]. The framework used imported scan images as the primary method for predicting Covid-19. The framework was based on multithreading from a variety of sensors related to the symptoms of Covid-19. In terms of category version, the accuracy of the Convolutional Neural Network models were compared. In [2] the authors proposed a model to predict mortality risk. The author provided more reasonable perceptions of the radiological and handling statistics that could be used in the Covid-19 prophecy prototype. They created an extrapolative forecast model based on the "XGBoost" machine-learning algorithm to estimate the patient's death. They devised a scientific path that simplified the process and assessed the rate of mortality.

The authors of the article [3] examined various models of Machine Learning (ML). The introspection and investigation indicated that ML models for coronavirus projection are a probability. The paper was entirely based on corona cases that have erupted in several countries. Scientific data were used to predict disease. On a variety of experimental projection tasks, the authors of [4] evaluated several algorithms, deep-learning approaches, and Intensive Care Unit counting techniques. The analysis was based on quantifiable datasets that were freely available to the general public. This assessment focused on the compatibility of various ML approaches and identifying appropriate ML techniques for projection.

Support Vector Machine and Artificial Neural Network-based systems, according to the authors S. Sukumaran and G. Kesavaraj, provided accurate prediction outcomes in E-Health Management services [5]. Tamilvizhi et al., used Random Forests and calibrated increased trees. Among the six algorithms compared, it provided the best version in all metrics. With massive datasets, Artificial Neural Networks have reached peak performance [6]. The algorithms heeded by Artificial Neural Networks were introduced by

Amit Ganatra and Hetal Bhavsar, with Logistic Regression having the highest precision. On the next pointer, SVM (Support Vector Machines) achieved the greatest precision [7].

Younus Ahmad Malla completed a clinical data set, and the report detailed the various efficient type algorithms (Diabetics). Seven data mining algorithms were tested, with Support Vector Machines outperforming Random Forests in terms of precision and quantitative measurements [8]. On the Wisconsin Prostate Cancer sources, Hajar Mousannif and Thomas Noel compared the accuracy and precision of various algorithms. All other data mining algorithms have been outperformed by SVM [9]. The article used a comprehensive benchmarking analysis to show that cognitive algorithms outperform other methods for envisioning tasks when a variety of scientific time-series data are used.

Amit Ganatra and Hetal Bhavsar conducted a comparison of classification algorithms to confirm that each algorithm will have its niche. They suggested that one use approaches for their data by evaluating the measurement system in use. An effective heart disease prediction system used Regression Techniques, Probabilistic Neural Networks, and Convolutional Neural to analyze the various advantages and disadvantages. When it comes to selecting mining objectives, the findings revealed that each approach has its selling point.

Hajar Mousannif and Thomas Noel compared the precision and accuracy of various algorithms on the Wisconsin Prostate Cancer sources. SVM has outperformed every other data mining algorithm. When a variety of research time-series data were used, the article used a thorough performance analysis to show that perceptual algorithms outperform other methodologies for envisioning tasks. Amit Ganatra and Hetal Bhavsar compared classification algorithms to ensure that each one has its niche. They recommended evaluating the measuring system in use before using approaches for their data. Regression Techniques, Probabilistic Neural Networks, and Convolutional Neural Networks were all used in an effective heart disease prediction system to weigh the benefits and drawbacks. The findings revealed that each approach has its selling point when it comes to selecting mining objectives.

Random Forests provide significant credible assumptions, according to Ya-Han Hu et al. [12]. Vanmathi et al., tested four data mining algorithms that were already familiar with the liver image database. The findings show that RF outperformed all other algorithms [13]. DT, Convolutional Neural Networks, SVM, and Linear Regression were particularly in comparison to generate prostate cancer resilience prediction models. Support Vector Machines and Convolutional Neural Networks [14] were found to be most accurate. M Fardaer et al., created a mortality rate forecasting standard using Classification Tree, Probabilistic Networks, and Convolutional Neural Networks. Deep Neural Networks were focused on providing precise classifications, according to the findings [15].

The Linear Regression model, Recurrent Neural Network (RNN), Bayesian Network, KNN, and Decision Trees have all been analogized in different metrics. The Logistic Regression sample, provided high accuracy reported by Neural Networks, as well as reliability [16]. These preliminary results showed that deciding which algorithm is the best is a time-consuming process. It applied classification algorithms to the field of crisis [17]. The work [18] investigated a wide range of algorithms, and Vector Support Machine and K-Nearest Neighbour were identified as the most accurate methodologies.

III. PROPOSED WORK

The research is performed in the Jupyter Console, which is a unified Python development and comprehension environment. The investigation is carried out in several stages, which are listed below. The data are in the form of an excel spreadsheet that contains the Covid cases from 2019 to 2021. First and foremost, Pandas is a Python-based software library for statistical manipulation and data analysis. The Pandas is also unrestricted software that was released under the BSD three-clause license.

A. Dataset

Taranvee disseminated the data used to train the prototype to recognize Corona. The dataset contains total cases, total deaths, new cases, ICU patients, total vaccinations, positivity rate, life expectancy, mortality cases, active Covid cases, hospitalized patients with Covid-19, and many other variables. Doctors' prescriptions for treating the disease were detailed in the authentic data. Those areas have been removed because this model does not require that information. A small portion of the data reveals whether the cases were previously associated with a specific illness, such as Cardiovascular Diseases, Gastrointestinal Diseases, and other illnesses with specific scientific values. There is a lot of textual data in this situation, as well as some exact values. Table I shows the facets of the dataset regarded for the predictive model.

TABLE I. UTILIZED FEATURES IN THE DATASET

Feature (Serial Number)	Feature (Name)
1	Continent
2	Country
3	Date
4	Total Cases

5	Total Deaths
6	Total Cases Per Million
7	Total Death Per Million
8	Reproduction Rate
9	Weekly Hospital Admissions
10	Population
11	Aged 65 Older
12	Aged 70 Older
13	GDP Per Capital
14	Cardiovascular Death Rate
15	Female Smokers
16	Male Smokers
17	Active Covid Cases

B. Data Pre-Processing

The outcome of the computational benchmark is determined by the data pre-processing step [20,21]. The gathered data are frequently handled haphazardly, with out-of-range significances, missing values, and other issues. As a result, data can mislead the outcome of the experiment. The superficial imputer of a sklearn python packet is used to generate skipped values in records. To fill in the missing values, the average operation is employed. Python package "SingleHotDecoder," which uses an ignoramus encoding strategy to regulate unambiguous data has been used.

C. Software Environment

Python is a well-known object-oriented programming language with lively semantics. Built-in data structures, combined with dynamic capturing and lively requirements, make it very easy for script to connect existing mechanisms. The readability of Python's simple, easy-to-learn syntax is prioritised, lowering the cost of programme maintenance. Python is concerned with "modules and packages," which increases code reuse.

The Python interpreter and a large traditional collection are made freely available in binary format for all major stages and can be distributed spontaneously. Pandas is still a popular Python data analysis and manipulation library. It comes with several functions and procedures that help speed up data analysis and pre-processing. Pandas are the subject of numerous reports and tutorials due to their popularity. NumPy arrays allow to perform advanced mathematical and other types of functions on massive amounts of data. Typically, such procedures are carried out more quickly and with less code than is possible with Python's built-in lines.

Matplotlib is a Python data visualisation and visualisation library that extends NumPy's statistical capabilities. As a result, it proposes an open-source alternative to MATLAB that is viable. To embed plots in GUI applications, developers can use Matplotlib's Application Programming Interfaces. Seaborn is an excellent Python visualisation library for data analysis and visualisation. It has a lot of default strategies and colour schemes to make statistical plots look better. Seaborn wants visualisation to be the most important aspect of data analysis and comprehension. It supports dataset-oriented APIs, allowing to switch between different visual representations of the same variables for a better understanding of the dataset. Tableau is a popular data visualisation tool in the business intelligence industry. It assists in the presentation of raw data in a format that is simple to comprehend. Tableau is needed to create data that can be accessed by employees at all levels of a company. Non-technical users can even make their own dashboards. Tableau is a data visualisation tool that quickly analyses data and creates dashboards and worksheets. People from all walks of life, including business owners, researchers, and those involved in a variety of endeavours, have become aware of the device.

D. Implementation

Panda's library is used to upload the attributes of the Covid-19 cases from 2019 to 2021 into the Jupyter Platform under Visual Studio Code (VSC) for data analysis, as well as to publish the data using the principal procedure in Python.

Continent	Country	Date	Total_Cases	New_Cases	Total_Deaths	New_Deaths	Total_Cases_Per_Million	New_Cases_Per_Million	Total_Deaths_Per_Million
Asia	Afghanistan	24-02-2020	5.0	5.0	NaN	NaN	0.126	0.126	NaN
Asia	Afghanistan	25-02-2020	5.0	0.0	NaN	NaN	0.126	0.000	NaN
Asia	Afghanistan	26-02-2020	5.0	0.0	NaN	NaN	0.126	0.000	NaN
Asia	Afghanistan	27-02-2020	5.0	0.0	NaN	NaN	0.126	0.000	NaN
Asia	Afghanistan	28-02-2020	5.0	0.0	NaN	NaN	0.126	0.000	NaN

Fig. 1. Imported Covid-19 Data from raw dataset

The read.csv command is used, along with the dataset's path to the Jupyter forum, which is already correlated to the Jupyter server as depicted in Fig. 1.

Population	Aged_65_Older	Aged_70_Older	GDP_Per_Capital	Cardiovasc_Death_Rate	Female_Smokers	Male_Smokers	Active_Covid_Cases
39835428.0	2.581	1.337	1803.987	597.029	NaN	NaN	5
39835428.0	2.581	1.337	1803.987	597.029	NaN	NaN	5
39835428.0	2.581	1.337	1803.987	597.029	NaN	NaN	5
39835428.0	2.581	1.337	1803.987	597.029	NaN	NaN	5
39835428.0	2.581	1.337	1803.987	597.029	NaN	NaN	5

Fig. 2. New Column added to the Dataset

In Fig. 2, a new column named "Active Covid Cases" has been added. This new column is added as the last queue in the whole data set. The column is evolved by deducting the Total Death Cases from the Total Cases. This formula subtracts all the death that ensued up to date from the total cases that appeared from the period of 2019 till 2021. Hence, the Total Active Cases lasting presently is computed [22].

To design a Pivot Table in Python, the Total Cases Value and Total Deaths Value, which can be used to find out the Mortality rate have been allocated. In this, a formula is employed i.e., "Total Deaths" is divided by the "Total Cases" that have happened so far and the divided value is then multiplied by 100% which assists to find the mortality rate. The mortality rate is sorted in descending order and the backdrop is themed in "gist-rainbow" from the color palette as shown in Fig. 3.

Country	Total_Cases	Total_Deaths	Mortality Rate
World	427815880.000000	5905942.000000	1.380487
High income	228787352.000000	2177438.000000	0.951730
Europe	153682248.000000	1695749.000000	1.103412
Upper middle income	116142099.000000	2418149.000000	2.082061
Asia	113619966.000000	1342138.000000	1.181252
European Union	105717953.000000	1001576.000000	0.947404
North America	92469988.000000	1362184.000000	1.473109
Lower middle income	81073459.000000	1268690.000000	1.564865
United States	78648651.000000	939064.000000	1.193999
South America	53491677.000000	1250853.000000	2.338407
India	42867031.000000	512622.000000	1.195842

Fig. 3. Mortality Rate

A bar plot is developed illustrating the leading Countries with the Significant Active Covid Cases. Firstly, the "Groupby" function is used to group all the values of the Covid Cases that have occurred till now and the above grouping is done based on the country as shown in the Fig. 4 and Fig.5. After that, the figure size and color are fetched into the code, and then the required values are cited to display the mandated output in the form of a Bar Plot.

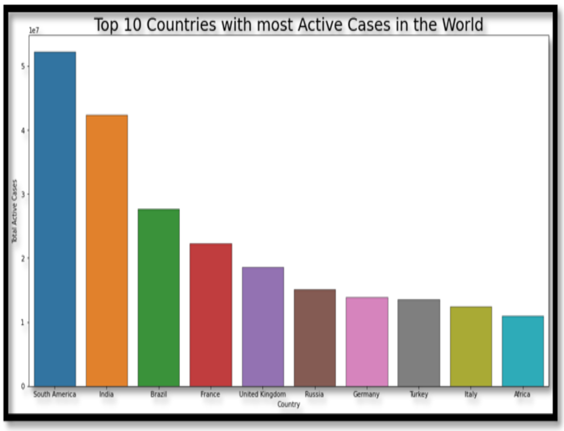

Fig. 4. Top 10 Countries with Most Active cases in the World

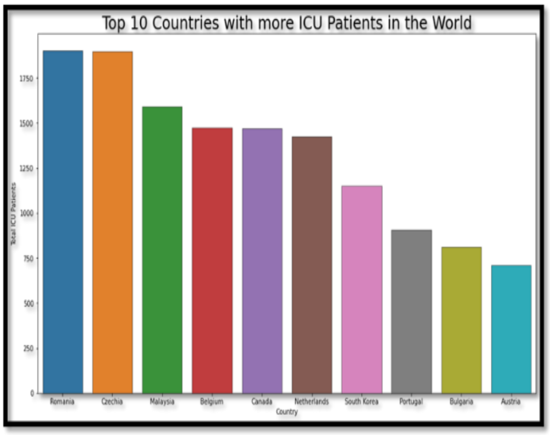

Fig. 5. Top 10 Countries with Most ICU Patients in the World

In Fig. 6, a pie chart is created for the total number of people vaccinated. The chart clearly explains the details of the vaccination such as Partially Vaccinated people and Fully Vaccinated People in the world. The chart is created by summing up the total partially vaccinated and fully vaccinated people's data and converting it into a Pie Chart.

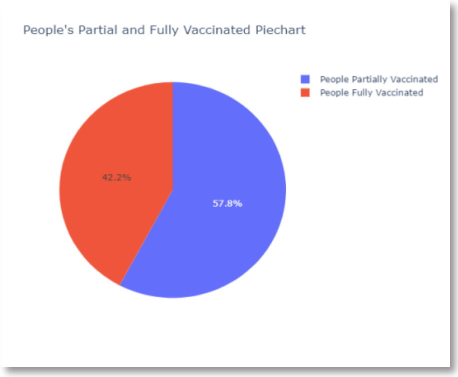

Fig. 6. Partially Vaccinated v/s Fully Vaccinated

Total vaccinations data is computed together to get the total amount of vaccinations done so far. This is then grouped by using the Country Value which is displayed in descending order in Fig. 7. The Bar Plot of Country wise vaccinations is shown in Fig. 8.

Location	Total_Vaccinations
European Union	1.698341e+11
United States	1.369127e+11
South America	1.299006e+11
Brazil	6.619190e+10
Africa	5.044346e+10
Germany	3.316337e+10
United Kingdom	3.235994e+10
Indonesia	3.023817e+10
Turkey	2.909759e+10
France	2.750630e+10

Fig. 7. Country wise Total Vaccinations

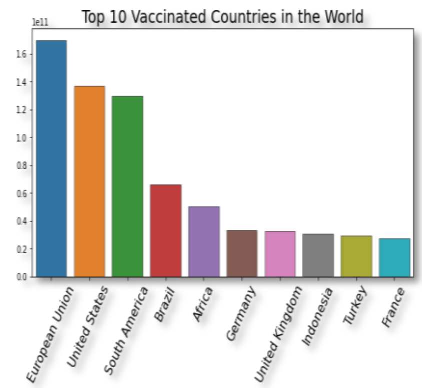

Fig. 8. Bar Plot of Country wise Vaccinations

Then the analysis is done for people aged more than 65 and 70 years. In this, the data of people aged more than 65 years and 70 years are grouped individually and then is summed up to get the total value in the entire world. This value is depicted in the form Pie Chart in Fig. 9.

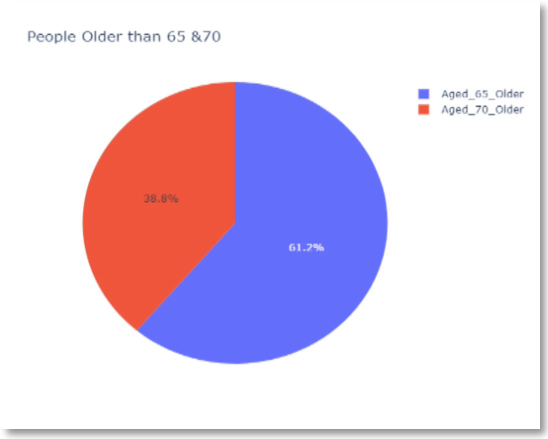

Fig. 9. People Aged more than 65 and 70 Years

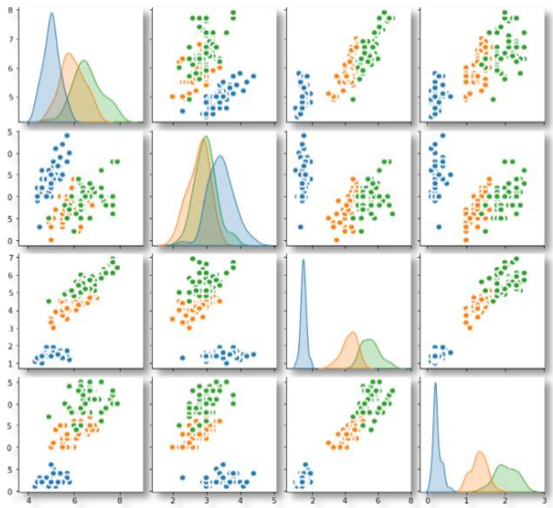

Fig. 10. Pair Plot illustration of Vaccination Data

In Fig. 10, the complete vaccination data is depicted in the shape of a Pair Plot. For this, the Matplotlib Library in Python is utilized. This library assists in plotting the vaccination data in diverse forms.

Total Deaths that happened due to cardiovascular diseases are depicted in Fig. 11, wherein Matplotlib library is used to show the Deaths data in the form of a Bar Plot. Initially, all the deaths that happened due to cardiovascular diseases are totaled and then the groupby function is used to group the total values. The total values are grouped in accordance with the Country Value. Then the required values are mentioned in the code and then finally the attributes are set to the Bar Plot and visualized.

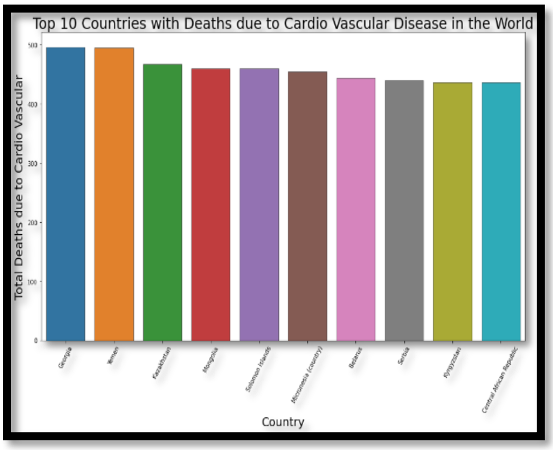

Fig. 11. Cardiovascular Deaths in the World

While scrutinising the patient's data pertaining to the period December 2019 to 2021, it is observed that 17.6% of the cases required admission into ICU. Further it is noticed that 82.4% of the cases was having mild symptoms and got treated as out patients. It is evident that these numbers are average by country to country, and location to location, which varied significantly, and shows the severity of the impact. It is shown in the Fig. 12. In Fig. 13 Total Vaccinations vs Total Boosters is shown.

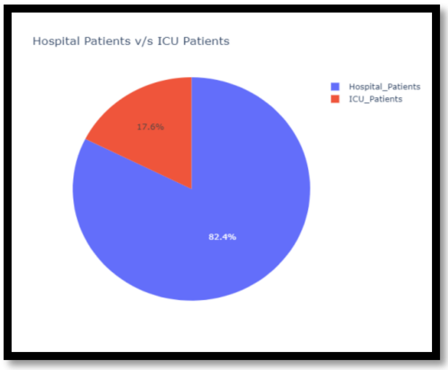

Fig. 12. Hospitalized Patient's Data

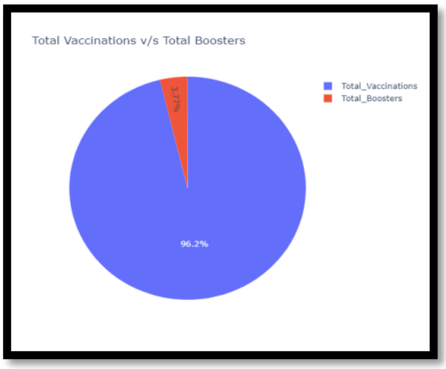

Fig. 13. Total Vaccinations vs Total Boosters

Dashboard of Covid-19 Data Analysis is displayed in Fig. 14.

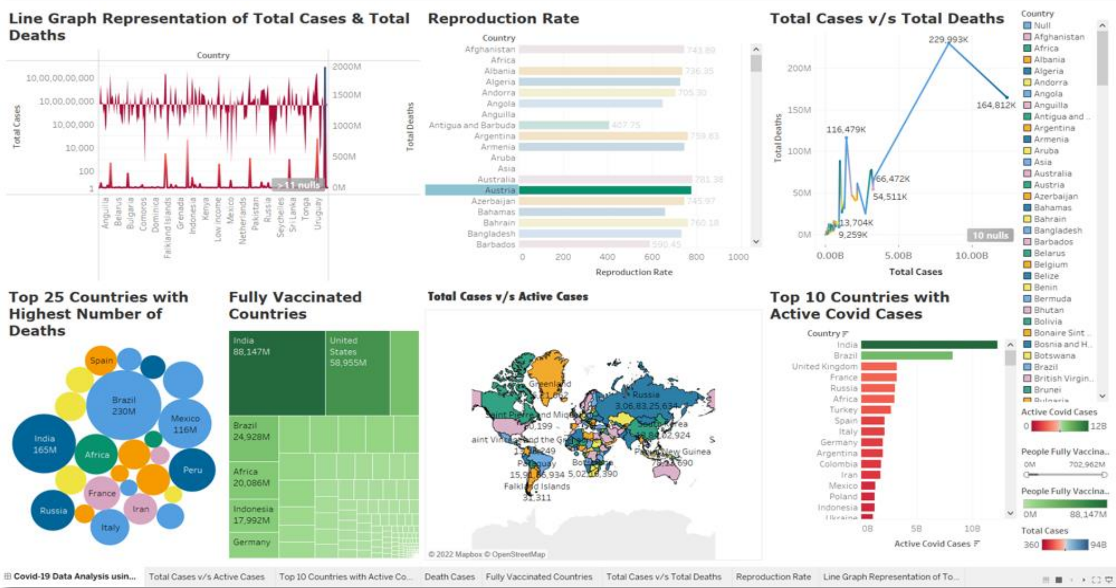

Fig. 14. Dashboard of Covid-19 Data Analysis

IV. CONCLUSION

The SEIR model is also being used to forecast the outbreak's path for the next 240 days, starting on January 20th, 2020. Since Covid-19 is still a transmissible disease with unknown or unknown properties, accurate SEIR predictions can only be made after the outbreak has been contained. Furthermore, bold projections of the widespread transformation tendencies in Korea, Iran, and Italy, reform measures, outbreaks and corresponding control times, as well as outlining countries' earliest communication dates, based on existing data from other countries have been made. Early in the crisis, evaluations of the model's elementary imitation value were found to be more lavish than one in all belongings, indicating increased contaminations of Covid-19 as predicted. The heretofore reported studies based on the sequence's intermission issuance of Covid are used to calculate the basic recurrence number using the log-linear model due to the lack of exact Covid-19 data for two locations.

ACKNOWLEDGMENTS

All authors have discussed the results and contributed to the final manuscript. The authors would like to thank Chennai Institute of Technology and Saveetha Institute of Medical and Technical Sciences for providing various resources and unconditional support for carrying out this research work.

References

[1] Halgurd S. Maghdid, et al. "A Novel AI-enabled Framework to Diagnose Coronavirus COVID 19 using Smartphone Embedded Sensors: Design Study," arXiv, 2020, https://doi.org/10.48550/arXiv.2003.07434

[2] Li Yan, et al. "Prediction of criticality in patients with severe Covid-19 infection using three clinical features: a machine learning-based prognostic model with clinical data in Wuhan," medRxiv, 2020, https://doi.org/10.1101/2020.02.27.20028027

[3] G. K. J, G. S, S. Rajendran, J. S. Vimali, J. Jabez and S. Srininvasulu, "Identification of Cyber Threats and Parsing of Data," 2021 5th International Conference on Trends in Electronics and Informatics (ICOEI), 2021, pp. 556-564, doi: 10.1109/ICOEI51242.2021.9452925.

[4] Ramesh Kumar Mojjada, Arvind Yadav, A.V. Prabhu, and Yuvaraj Natarajanc, "Machine learning models for covid-19 future forecasting," Mater Today Proc. 2020, Dec 9, doi: 10.1016/j.matpr.2020.10.962.

[5] S. Sukumaran and G. Kesavaraj, "A study of data mining classification techniques," Fourth International Conference on Computing, Communications, and Networking Technologies (ICCCNT), pp.1–7, 2013

[6] R. Surendran and T. Tamilvizhi, "Cloud of medical things (CoMT) based smart healthcare framework for resource allocation," 3rd Smart Cities Symposium (SCS 2020), 2020, pp. 29-34, doi: 10.1049/icp.2021.0855.

[7] Amit Ganatra and Hetal Bhavsar, "A comparison of supervised machine learning training algorithms," Vol.2 (4), pp.2231–2307, "International Journal on Computing and Engineering (IJSCE)", 2012.

[8] Malla, Younus Ahmad, "A machine learning approach to predicting breast cancer in its early stages," The International Journal of Engineering and Computer Science, 2017.

[9] Hajar Mousannif, and Thomas Noel, "Machine learning algorithms are being used to predict and diagnose breast cancer risk," Procedia Computer Science, Vol. 83, pp.1064–1069, 2016.

[10] R. Surendran and T. Tamilvizhi, "Friendly Online Technology Development Cloud Service for Bahraini Students based on E-Advisor," 2018 International Conference on Innovation and Intelligence for Informatics, Computing, and Technologies (3ICT), 2018, pp. 1-7, doi: 10.1109/3ICT.2018.8855783

[11] Tamilvizhi T, Parvatha Varthini B. and R. Surendran (2015), "An improved solution for resource management based on elastic cloud balancing and job shop scheduling," ARPN Journal of Engineering and Applied Sciences, ISSN 1819-6608 (Online), Vol. 10, no. 18.

[12] Ya-Han Hu, Feng Kang, and Pei-Ju Lee, "Inpatient pressure wound prediction models are being built using machine learning techniques," CIN 2020 stands for Computers, Informatics, and Nursing, 2020.

[13] C. Vanmathi, M. Ramaiah, P. Baranwal, S. B. Shastri, M. Vanitha, and M. Ramaiah, "Machine learning techniques are compared analytically for a liver dataset," Vol.1, 2019 Innovations in Power and Advanced Computing Technologies (i-PACT), pp.1–5, 2019.

[14] M Fardaer, Narges Alizadeh Noohi, and Marzieh Ahmadzadeh. "Medical data mining and a predictive model for colon cancer survival are being developed," International Journal of Innovative Research in Engineering and Science, Vol. 2(2), 2013.

[15] S. Sukumaran and G. Kesavaraj, "A study of data mining classification techniques," Fourth International Conference on Computing, Communications, and Networking Technologies (ICCCNT), pp.1–7, 2013.

[16] T. Tamilvizhi and R. Surendran, "Time and cost-effective recovery mechanism for unhealthy resources in proactive fault tolerance framework," 3rd Smart Cities Symposium (SCS 2020), 2020, pp. 444-449, doi: 10.1049/icp.2021.0756.

[17] Surendra Babu, Bendi Venkata Ramana, Venkateswarlu, "A critical examination of classification algorithms in the diagnosis of liver disease," International Journal of Database Management Systems, Vol. 3, no. 2, pp.101–114, 2011.

[18] Tamilvizhi T and Parvatha Varthini.B, "Online vaccines and immunizations service based on resource management techniques in cloud computing," BIOMEDICAL RESEARCH-INDIA; Special Issue, Special Section:Health Science and Bio Convergence Technology : Edition-I, pp. S392-S399, 2016

[19] Sungheetha and Akey. "COVID-19 Risk Minimization Decision Making Strategy Using Data-Driven Model," Journal of Information Technology 3, no. 01, pp. 57-66, 2021.

[20] Dhaya, R. "Deep net model for detection of covid-19 using radiographs based on ROC analysis," Journal of Innovative Image Processing (JIIP), vol. 2, no. 03, pp. 135-140, 2020.

[21] Ali, B. Alshurooqi and R. Surendran, "Mobile-based pressure sore prediction and prevention system (PSPPs)," 3rd Smart Cities Symposium (SCS 2020), 2020, pp. 415-420, doi: 10.1049/icp.2021.0888.

[22] Tamilvizhi.T, Surendran.R, Bommi. R. M, "Radio Frequency Identification (RFID) based ubiquitous health care data handling", IOP Conference Series: Materials Science and Engineering, 2020, 994(1), 012021.